

A dual-branch siamese spatial-spectral transformer attention network for Hyperspectral Image Change Detection

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ABSTRACT

The convolutional neural networks have recently gained widespread attention in Hyperspectral Image Change Detection (HSI-CD) due to their outstanding feature extraction ability. However, limited by the inherent network backbones, the convolutional neural networks (CNNs) fail to mine the sequence attributes and model the intricate relationships of spectral signatures. In contrast, transformers are proficient at learning sequence information owing to the powerful self-attention mechanisms. The two backbone structures exhibit complementary spatial and spectral feature extraction strengths, respectively. Inspired by this, we propose a dual-branch siamese spatial-spectral transformer attention network (DBS³TAN) for HSI-CD. The main idea is to fully exploit the advantages of CNNs and transformers for spatial and spectral feature extraction. More importantly, we devise the two key modules, i.e., the spatial attention module and the spatial-spectral transformer module. The former utilizes depthwise separable convolutions and attention mechanisms to emphasize the features of dual-temporal HSIs from the spatial perspective. The latter focuses on the sequence attributes of spectral signatures and mines the spatial characteristics from adjacent pixels. We employ the weighted contrastive loss function to separate the changed and unchanged pixels more reliably and set the random weight factors to balance the contributions of the two branches. Finally, the threshold values judgment is used to obtain the ultimate detection maps. We conduct extensive experiments to evaluate the DBS³TAN on three HSI datasets, demonstrating its superior performances than compared methods qualitatively and quantitatively. The source code will be available at <https://github.com/zhangyiyan001/DBS3TAN>.

1. Introduction

With the development of earth observation technology, hyperspectral remote sensing gained much attention and has been widely applied in global earth monitoring (Guo, Zhang, & Zhu, 2015). Hyperspectral imaging technology samples with small gaps over a wide range of spectral wavelengths by the spectrometer, making it possible to detect and classify land covers at a finer scale (ElMasry & Sun, 2010). Different from the general remote sensing images such as optical images (Tewkesbury, Comber, Tate, Lamb, & Fisher, 2015), light detection radar (LiDAR) (Behroozpour, Sandborn, Wu, & Boser, 2017), Synthetic Aperture Radar (SAR) (Freeman, 1992) and Multispectral images (Chaudhuri & Samal, 2008). Hyperspectral images (HSIs) are characterized as hundreds of spectral bands and high spectral resolution, which contain rich information about land covers and can detect substances with diagnostic characteristics (Hong, Gao, Yokoya, et al., 2021). Consequently, HSIs play an essential role in the intelligent

interpretation of remote sensing (RS), i.e., image classification (Gao, Zhang, Chen, & Li, 2021; Hong, Gao, Yao, et al., 2021; Hong, Han, et al., 2021; Su et al., 2022), semantic segmentation (Hong, Yao, Meng, Xu, & Chanussot, 2020; Pan et al., 2020), and object detection (Gao et al., 2023; Li, Zhang, Zhang, Chen, & Gao, 2023; Pham et al., 2019).

Hyperspectral Image Change Detection (HSI-CD) is a crucial application intended to identify change areas pixel-to-pixel from bi-temporal HSIs in the same geographic area, which has forward-looking implications for the applications, e.g., precision agriculture (Caballero, Calvini, & Amigo, 2019), military reconnaissance (Longbotham et al., 2012), and flood monitoring (Briottet et al., 2006). As a result, HSI-CD has become an active topic in the RS community.

Many traditional CD methods can be broadly classified into three categories. The first is based on simple algebraic operations. Malila (1980) employed calculating spectral change vectors from bi-temporal images named Change Vector Analysis (CVA) to detect the changed and

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unchanged areas. Based on this work, Liu, Bruzzone, Bovolo, and Du (2014) introduced a novel Sequential Spectral Change Vector Analysis (S2CVA) for HSI-CD, which utilized an adaptive reference vector scheme to drive the change representation by following a top-down structure to exploit the multiple change information hierarchically and automatically. However, the algebraic-based approaches are likely to suffer the problem of information redundancy, leading to poor detection performance. The second category was image classification methods, which regarded CD as binary-class classification tasks and utilized classifiers to separate the changed and unchanged areas. Nemmour and Chibani (2006) investigated the supervised method, i.e., Multiple Support Vector Machine (SVM) for HSI-CD, which produced output through the decision function. The pixel will be assigned to the change class if the function is positive. Otherwise, it will be classified as the unchanged class. The third category was image transformation methods, including PCA (Li & Yeh, 1998), MAD (Nielsen, Conradsen, & Simpson, 1998), and SFA (Wu, Du, & Zhang, 2013), which transformed HSIs into the new feature space for dimension reduction and emphasize the changed features.

Recently, deep learning (DL) has become a hotspot in the field of RS image processing. Unlike traditional methods, DL-based CD methods can automatically explore spatial-spectral features with end-to-end network layers due to the powerful feature representation ability. Various deep learning-based networks, such as deep belief networks (DBNs) (Huang, Yu, & Feng, 2019), deep neural networks (DNNs) (Du, Ru, Wu, & Zhang, 2019), and autoencoders (AEs) (Hu, Wu, Zhang, & Du, 2021), have been applied to hyperspectral image classification and detection (HSI-CD) and have demonstrated effectiveness. However, they primarily focus on spectral feature extraction. Convolutional Neural Networks (CNNs) (Li, Liu, Yang, Peng, & Zhou, 2021) are representative backbones of DL, characterized by excellent feature extraction ability for 2D images. CNN-based models have been widely used in HSI-CD. Wang, Yuan, Qian, and Li (2018) introduced a mixed-affinity matrix for enhanced feature mining and information fusion, along with 2D CNNs for effective feature learning and improved generalization ability. Wang, Xing, Feng, Song, and Mu (2021) first decomposed the bi-temporal HSIs into directional subbands using the 3D-Wavelet transform. Then an active learning strategy is iteratively employed to extract deep features from the constructed feature maps. Similarly, Liu et al. utilized spatial information and active learning to select highly reliable unlabeled samples to increase the training set, facilitating the complete semi-supervised HSI-CD. Ou, Liu, Tu, Zhang, and Xu (2022) presented a CNN framework involving slow-fast band selection and feature fusion grouping, which incorporated the slow-fast band selection and feature fusion grouping to generate a more discriminative feature group, obtaining competitive results on HSI-CD.

Due to the fact that HSIs contain both spatial and spectral information, it is essential to make good use of these features to improve CD performances. Zhan, Song, Sun, Jia, and Wu (2020) proposed a hybrid network that integrated 1D and 2D CNNs to extract discriminative spectral-spatial features jointly and simultaneously accelerate network convergence. Wang, Wang, Wang, and Bruzzone (2022) proposed an end-to-end residual self-calibrated network (RSCNet), which adaptively built inter-spatial and inter-spectral dependencies around each spatial location with fewer extra parameters and reduced complexity. Hu, Wu, and Zhang (2022) introduced a novel pixel-level self-supervised hyperspectral spatial-spectral feature understanding network for accurate HSI-CD. By considering the entire images instead of patches and utilizing a spatial-spectral attention module, HyperNet effectively captures spatial correlation and discriminative spectral features. Zhan, Song, Xu, Wan, and Wu (2021) devised a spectral-spatial convolutional neural network with siamese architecture that employed the Euclidean distances to represent the similarity between spectral-spatial tensor pairs.

Although these methods achieve robust performance compared with traditional methods, they tend to suffer from information redundancy

and computational costs in the convergence process. Attention mechanisms have been widely used and introduced into the HSI-CD to improve these problems. Yang et al. (2022) proposed a deep multiscale pyramid network enhanced with spatial-spectral residual attention for HSI-CD, which has a solid capability to mine multilevel and multiscale spatial and spectral features, improving the performance in complex changed regions. Li, Ren, et al. (2023) proposed a cross-band 2D self-attention network (CBANet) for end-to-end HSI-CD. By embedding a cross-band feature extraction module into a 2-D spatial-spectral self-attention module, CBANet can extract the spectral difference of matching pixels by considering the correlation between adjacent pixels. To focus more on the bi-temporal information, Song, Ni, Cheng, and Wang (2022) introduced an interaction symmetric attention network to utilize cross-temporal attention to enhance the feature representation ability, extracting the change information in a symmetric pattern. Gong et al. (2021) designed a spectral and spatial attention network capable of suppressing CD-irrelevant spectral and spatial information via adaptive spectral and spatial attention mechanisms. Qu, Hou, Dong, Li, and Xie (2021) introduced a multilevel encoder-decoder network that employed a contextual-information-guided attention module to yield useful spatial-spectral features and improve detection performance. However, the above-mentioned attention-based methods fail to highlight the differences between bi-temporal images in HSI-CD, leading to deficiencies in judging the discrepancy regions. Wang, Wang, Wang, and Atkinson (2021) introduced an end-to-end model that integrated the CBAM attention mechanism into the siamese network, which can emphasize informative channels and locations and adopt more reliable solutions to separate changed and unchanged pixels.

Additionally, to obtain the complete spectral sequence information (Dosovitskiy et al., 2020; Vaswani et al., 2017), transformer-based models have been introduced in HSI-CD. Wang, Hong, et al. (2022) designed a spectral-spatial-temporal transformer network, which encoded each pixel to remember spectral and spatial sequences and utilized a class token to store the corresponding class information. Similarly, Ding, Li, and Zhao (2022) employed pixel embedding module to guide transformers to exploit change information of space and time. Then the multi-head attention mechanism was utilized to learn several attentions and extracts the joint-weighted spatial-spectral-temporal features. Wang, Zhao, Zhao, and Li (2023) proposed a triplet transformer framework for HSI-CD. The model first used self-attention to capture the correlation between arbitrarily two pixels in the same patch and then constructed a novel of parents-temporal attention to account for the additional temporal image. Shafique, Seydi, Alipour-Fard, Cao, and Yang (2023) introduced a hybrid spatial-spectral convolutional vision transformer (SSViT) for HSI-CD to improve the performances. By combining the strengths of CNNs and transformers, the SSViT effectively models global and local dependencies, achieving significantly improved performance compared to existing HSI-CD methods. To handle the limited training samples problems, Dong, Yang, Qu, Xiao, and Li (2023) presented a dual-branch local information-enhanced graph-transformer (D-LIEG) network. The model achieved promising performances under the challenge of limited training samples via combining local and global spectral-spatial features through a cascaded LIEG block and a novel graph transformer.

However, the above-mentioned attention-based methods fail to highlight the differences between bi-temporal images in HSI-CD, leading to deficiencies in judging the discrepancy regions. Although these methods have obtained promising detection results, they still have difficulty capturing subtle discrepancies along spectral dimensions, leaving room for improvement in detection performance. We aim to develop a dual-branch siamese spatial-spectral attention transformer network (DBS³TAN) for HSI-CD. The main contributions of this paper can be summarized as follows.

- (1) The main backbone of DBS³TAN adopts the dual-branches siamese network, significantly reducing the computational complexity owing to the weight-sharing strategy. The dual-branch

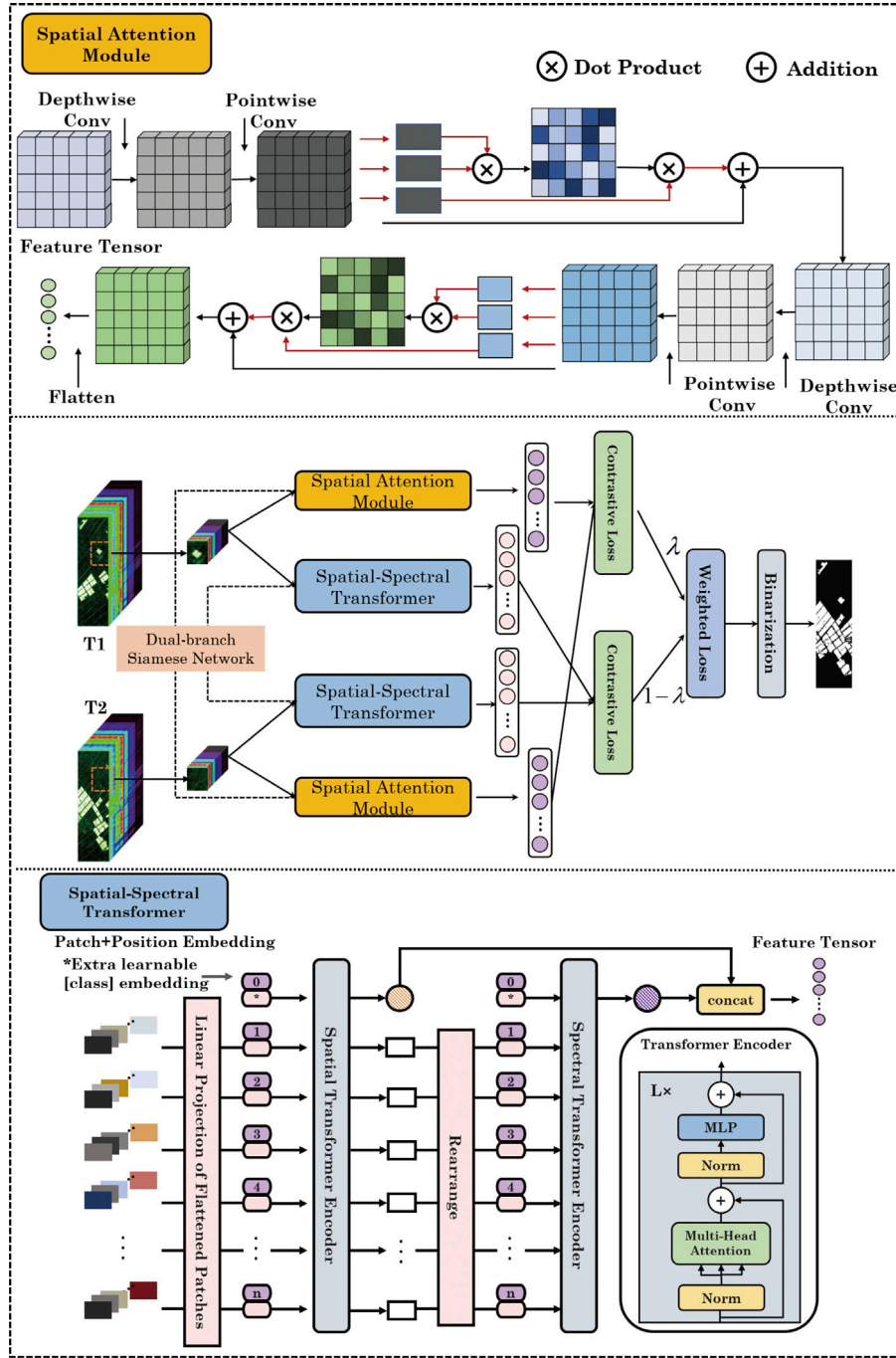


Fig. 1. The diagrams of the proposed model and corresponding modules. From top to bottom: the spatial attention module; the framework of the proposed DBS³TAN; the spatial-spectral transformer module.

siamese network not only preserves the original information but also highlights the discrepancies between bi-temporal HSIs.

- (2) Considering CNNs only specialize in capturing contextual characteristics in the spatial domain while ignoring long-term dependencies of spectral signatures, we introduce the transformer and adopt the two-branch network. One branch consists of the spatial attention module, aiming to highlight the discriminative spatial features. The other branch comprises the spectral-spatial transformers module, which is intended to model the pixel-wise relationships and sequence attributes of spectral signatures from the global perspective.
- (3) We refined the weighted contractive loss function and made it more applicable to converge the network, alleviating the

influences of imbalanced class samples. Furthermore, the progressive loss function encourages the feature vectors of changed pixels to be far apart from each other while keeping the vectors of unchanged pixel pairs close, i.e., increasing the inter-class separability between changed and unchanged pixel pairs. We adopt the different weights factor to balance the contributions between spatial and spectral branches to achieve better fusion performances.

- (4) We conducted extensive experiments to evaluate the detection performance of the proposed DBS³TAN on three HSI datasets. The experimental results demonstrate that our proposed model outperforms other state-of-the-art backbone networks.

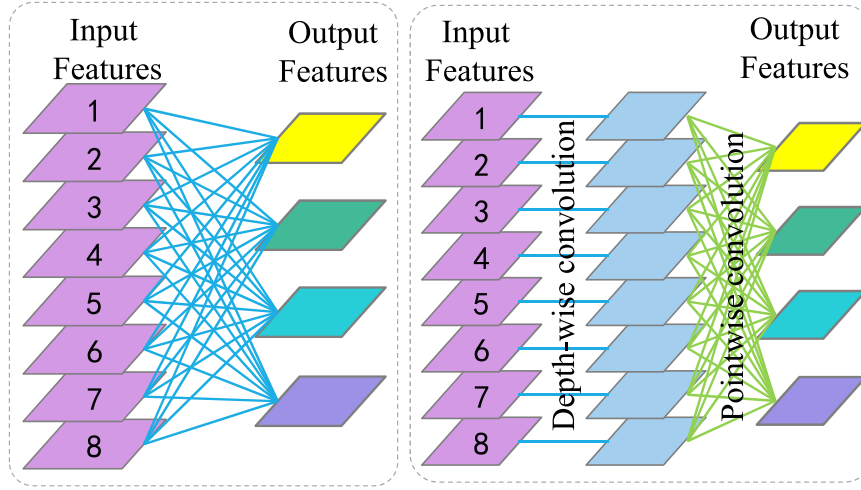


Fig. 2. The diagrams of the proposed model and corresponding modules. From top to bottom: the spatial attention module; the framework of the proposed DBS³TAN; the spatial-spectral transformer module.

The paper is organized as follows. Section 1 reviews the basic knowledge of transformers. The proposed methodology is elaborated in Section 2. Extensive experiments with ablation studies and discussions are conducted in Section 3. The conclusion is summarized in Section 4.

2. Methodology

2.1. Overview of the proposed DBS³TAN

We aim to design a novel dual-branch siamese spatial-spectral transformer attention network to capture the discrepancies between the dual-temporal HSIs, making it well-applicable for accurate change detection of land covers. Overall, we devise two individual modules, i.e., spatial attention module and spatial-spectral transformer module, integrating them into the two branches of the siamese network to precisely distinguish the variations from the dimension of space and spectrum. Moreover, the proposed transformer module improves the detail-capturing capacity of subtle spectral discrepancies and enhances information transitivity. The parameters are shared in the siamese network to reduce the calculation cost so that the dual-branch networks are equipped with the same feature extraction ability, allowing the discovery of the differences between the dual-temporal images. Finally, the refined contrastive loss is adopted to obtain the final detection maps. Fig. 1 illustrates the diagrams of the proposed DBS³TAN and the corresponding modules in HSI-CD tasks.

2.2. Spatialwise attention module

As it is widely recognized, the standard 2D CNN architecture is highly effective in extracting spatial features due to the topology of the convolution kernels. Nevertheless, it often incurs more computational costs and longer training times. To address the problem, we propose employing depthwise separable convolutions (DSC) to replace ordinary convolutions. Fig. 2 provides an illustrative diagram of ordinary convolution and depthwise separable convolution (DSC) (Chollet, 2017). The key difference between the two types of convolutions is that the ordinary 2D convolution considers all channels of the corresponding region simultaneously. In contrast, the DSC integrates all extracted spatial features and employs 1×1 convolutions to learn channel correlation information. Then, the feature maps are sent to the attention module to emphasize the discriminative spatial features. Specifically, given the input data $X \in R^{C \times H \times W}$, where C represents the channel number, and H and W represent the height and width of the feature maps, respectively. The input feature F clones itself to obtain three sets of

features $\{F_Q, F_K, F_V\} \in R^{C \times H \times W}$. Then, F_H and F_W are transformed into space $R^{C \times N}$, where $N = H \times W$. The spatial attention matrix F_{spa} is obtained through matrix multiplication operations between the transpose matrix of F_Q and F_K using the softmax operation:

$$F_{spaX,ij} = \frac{\exp(F_{Q,i} \cdot F_{K,j}^T)}{\sum_{k=1}^N \exp(F_{Q,k} \cdot F_{K,j}^T)} \quad (1)$$

where $F_{spaX,i}$ represents the impact of position i on the feature of the central position j , and $F_{spaX,i}$ indicates the pairwise correlation between features. The final output spatial attention $F_{spaA,j}$ can be denoted as:

$$F_{spaA,j} = \sum_{i=1}^N (F_{spaX,ij} \cdot F_{V,i}) + F_j \quad (2)$$

The spatial attention module selectively improves the compactness and semantic consistency within the class, allowing the network to identify the changing state of the land covers in different temporals accurately. We repeated the depthwise separable convolutions and attention module twice to enhance the land-cover feature representation ability.

2.3. Spatial-spectral transformer

Existing transformer structures mainly focus on extracting nonlinear spectral features from the perspective of sequential representations, achieving promising performances in practical applications. However, the spatial relationship between adjacent pixels significantly affects the improvement of CD task performances. As a result, we reconsider whether the transformer structure can obtain sequential spectral representations and spatial adjacent pixels information simultaneously.

At the preprocessing stage, bi-temporal HSIs are represented as X_{t1} and X_{t2} with a spatial resolution of $H \times W$ and a channel number of C . We construct multiple patches by combining the central pixels, producing patches with a size of $S \times S \times C$, where S is the length and width of the image patches. We consider patch pairs to represent the two-temporal HSIs, $X = \{X_1, X_2\}$. The standard transformers receive a 1D sequence of token embeddings as input. To handle 3D HSIs, we reshape the image into a sequence of flattened 2D patches. Specifically, for the first temporal branch, the patches are reshaped into the spectral sequence set $x = x_1, x_2, x_3, \dots, x_c$. The resolution of the original image (e.g., (H, W)) has been compressed into 1D spectral attributes. The height and width of image patches can be denoted as h and w , and also chosen as 1, which enables us to regard each image patch as an individual pixel. This approach is entirely different from how the standard

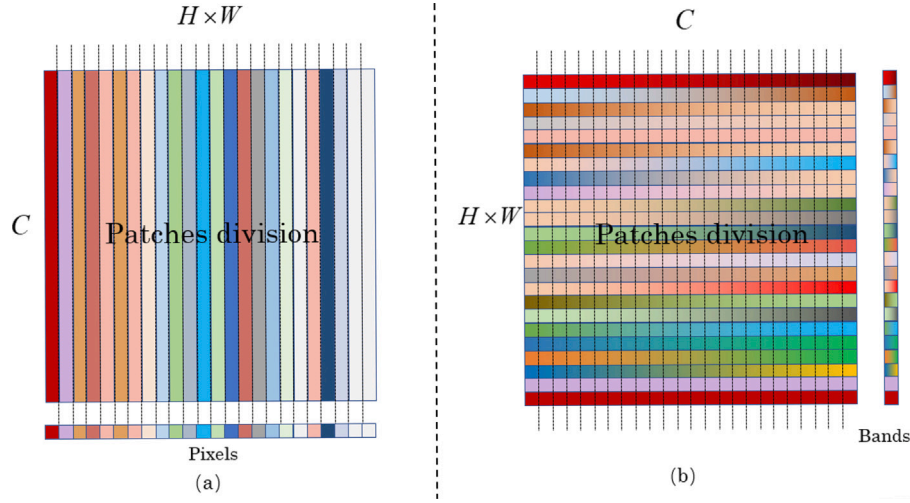


Fig. 3. The diagram illustrations of two kinds of patches division. (a) for spatial transformer (b) for spectral transformer.

transformer (i.e., Spectralformer) operates. Fig. 3 illustrates the two types of methods used to divide the image patches. The sizes of image patches impact the detection performances on three datasets, which will be discussed in detail in Section 3. In brief, mainstream methods (e.g. Spectralformer) divide patches along the spectral bands to explore the differences in spectral sequences from the hundreds of bands to better learn the relationship between spectral channels. In contrast, our proposed division method focuses more on the individual and its adjacent pixels, which significantly adapts the transformer structure to apply the spectral sequences to the spatial structures. In other words, we switched the viewpoint and fed the sequence of individual pixels as data into the spatial transformer. With the subsequent MSA, the model can capture abstract and high-level spatial features of HSIs. Similar to the standard transformer, we also employ a learnable embedding for the sequence of embedded patches, which is randomly initialized and can be regarded as the abstract image representation to learn the spatial features as the network converges, which can be regarded as

$$\hat{X} = \text{concat}(X_{cls}, X_{patch}) \quad (3)$$

where X_{patch} represents the multiple image patches and X_{cls} is the random initialized class tokens. Then we regard multiple small image patches x as a matrix of $wh \times c$ for the convenience of calculation, where c , w and h are the channel, width and height of the image patches, respectively. The n dimensions constant latent vectors obtained by linear projection are formulated by

$$E = w_1 x \quad w_1 \in \mathbb{R}^{n \times wh} \quad (4)$$

where w_1 is a trainable weight matrix with size $n \times wh$ and $E \in \mathbb{R}^{n \times c}$ covers the linear projection features of all spectral bands. n represents the constant latent dimensions. Although each temporal HSI is divided into lots of patches, the location of each image patch has specific contextual information and is of great significance for the accuracy of HSI-CD. As a result, position embedding is also introduced to mark the location of each image patch. The position embedding p is combined with the vector E by

$$Z_0 = E + p \quad E \in \mathbb{R}^{n \times c}, p \in \mathbb{R}^{n \times c} \quad (5)$$

where p represents a random matrix with the size of $n \times c$ and is continually changing and converging to the optimal values with the model training, then the Z_0 is fed into the transformer encoder composed of multiple layers of MSA and MLP for encoding the spectral sequences. The formula can be described as follows:

$$\begin{aligned} Z'_d &= \text{MSA}(\text{LN}(Z_{d-1})) + Z_{d-1}, \quad d \in \{1, 2\} \\ Z_d &= \text{MLP}(\text{LN}(Z'_d)) + Z'_d, \quad d \in \{1, 2\} \end{aligned} \quad (6)$$

where LN represents the layernorm operation, which is used to regulate the optimization space and ensure the stability of data distributions. The Z_d is the output of the encoder, and the number of the encoder is set to 2. Z_2 is the final output of the transformer encoder.

Then, we rearrange the shape of output from $Z_2 \in \mathbb{R}^{n \times c}$ into $Z'_2 \in \mathbb{R}^{c \times n}$. The following operations can be regarded as the spectral transformer, extracting sequences discrepancies along the spectral channels like spectralformer. Similar to the previous spatial transformer, an extra learnable class embedding $X'_{cls} \in \mathbb{R}^{1 \times n}$ is also prepended for image representations, which can learn pseudo-category of patches through the transformer encoders. The $\hat{X}' \in \mathbb{R}^{(c+1) \times n}$ can be concatenated by the X'_{cls} and Z'_2 .

$$\hat{X}' = \text{concat}(X'_{cls}, Z'_2) \quad (7)$$

Next, we feed $\hat{X}'$ into the spectral transformer encoder, which has the same structure as the spatial transformer encoder for modeling the spectral sequence features. After the operations mentioned above, the separated two class tokens can encode the statistical characteristics of HSIs from both the spatial and spectral perspectives, respectively, which can be regarded as the aggregations of semantic textual information. As a result, we concatenate the two class tokens to form the comprehensive class token vector. It can be formulated as

$$\bar{X}_{cls} = \text{concat}(X_{cls}, X'_{cls}) \quad (8)$$

where $\bar{X}_{cls}$ is the final concatenated class token, which is fulfilled with high-level and abstract contextual features from both spatial and spectral dimensions, facilitating the loss calculation.

2.4. Contrastive loss function

The selection of an appropriate loss function is essential in achieving optimal network convergence and obtaining superior change detection performance. The triplet loss is commonly used in the siamese network, which aims to minimize the distance between the anchor and the positive example while maximizing the distance between the anchor and the negative example. However, it is more applicable for face recognition. In comparison, the contrastive loss function represents a specific class of loss functions utilized for measuring the similarity between two sets of samples. It consists of two or more identical neural networks trained in parallel to learn a model that scores the similarity between inputs, making it an ideal candidate for addressing the problem of HSI-CD.

Notably, the effectiveness of the contrastive loss function in change detection tasks depends on the extent to which the feature vectors of

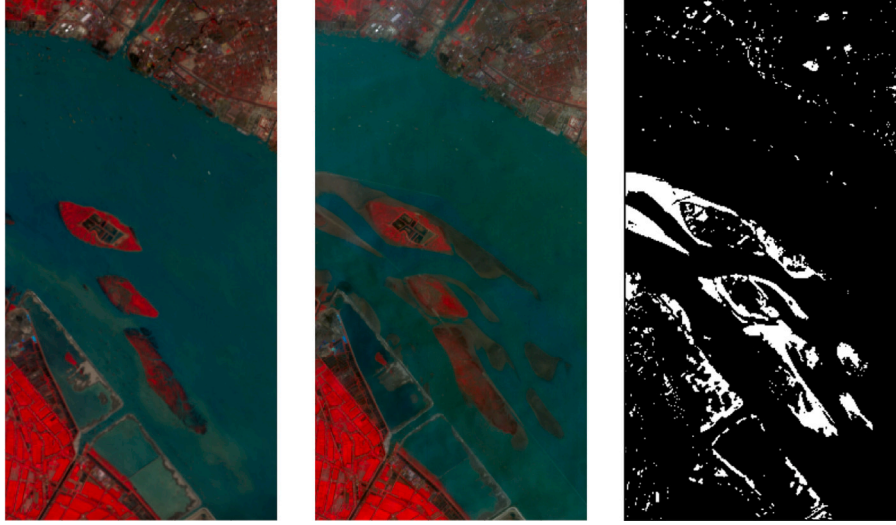


Fig. 4. River dataset. From left to right: Image acquired on May 3, 2013; Image acquired on December 31, 2013; Ground Truth.

unchanged pixel pairs are closely positioned while those of changed pixel pairs are distant from one another. To this end, given bitemporal patches X_1 and X_2 that cover the same center pixel $x_{(i,j)}$, the output feature tensor of the network for the center pixel $G(X)_{i,j} \in \mathbb{R}^{1 \times N}$ can be used to compute the Euclidean distance between the feature vectors $G(X_1)_{i,j}$ and $G(X_2)_{i,j}$. Specifically, the Euclidean distance serves as a means of measuring the gap between the feature vectors, and it is expressed as follows:

$$D(X_1, X_2) = \|G(X_1)_{i,j} - G(X_2)_{i,j}\|_2 \quad (9)$$

And the loss function can be expressed as follows:

$$\begin{aligned} \text{Loss}(w) &= \sum_{k=1}^p l(W, (y, X_1, X_2)^k) \\ &= \sum_{k=1}^p \left((1 - y_{i,j}^k) l_U(D_{i,j}^k) + y_{i,j}^k l_C(D_{i,j}^k) \right) \end{aligned} \quad (10)$$

The label y denotes the binary classification of patch pairs X_1 and X_2 , with a value of 0 assigned to unchanged pairs and 1 to changed pairs, represented as $y(i, j)$ in the notation. The $(y, X_1, X_2)^k$ consists of k labeled patch pairs, each with a corresponding label y . The loss function $l_U(D_{i,j}^k)$ and $l_C(D_{i,j}^k)$ are specifically designed to measure the loss for the unchanged and changed pixel pairs, respectively. In particular, $D_{i,j}^k$ represents the Euclidean distance between the feature vectors of the two patch pairs, $G(X_1)_{i,j}$ and $G(X_2)_{i,j}$. The formula for the Contrastive Loss Function can thus be expressed as follows:

$$l_U(D_{i,j}^k) = \left(\sigma(D_{i,j}^k) \right)^2 \quad (11)$$

$$l_C(D_{i,j}^k) = \max \left\{ 0, q - \left(\sigma(D_{i,j}^k) \right)^2 \right\} \quad (12)$$

where q represents the margin and is empirically set to 1, increasing the sigmoid function's value for changed pixel pairs, denoted as $\sigma(D_{i,j}^k)$, toward 1. Considering the different contributions of the two branches, the CD accuracies are likely to be influenced. In order to obtain the best detection results, it is necessary to incorporate a weighted factor, which can be characterized as follows:

$$\text{Loss}_{total} = \lambda \text{Loss}_{spa} + (1 - \lambda) \text{Loss}_{spe} \quad (13)$$

where Loss_{total} is the final loss and λ is the weighted factor to balance the contributions between the spatial attention branch and spatial-spectral transformer branch. After performing the aforementioned operations, the final detection images for the three datasets are obtained by comparing each pixel value with the pre-set threshold value. The

pixels with values above the threshold are classified as changed pixel, other are assigned as unchanged pixel.

3. Experiments

In this section, we first introduce three publicly available HSI datasets used for change detection. Subsequently, we describe the detailed experimental setup, including evaluation metrics, comparative backbone networks, and implementation details. Finally, we conduct extensive experiments, including hyperparameter sentiment analysis and ablation studies, to evaluate the effectiveness of the proposed DBS³TAN model.

3.1. Experimental datasets

The River dataset was acquired by the Hyperion sensor on May 3, 2013, and December 31, 2013, which recorded the change of a river in Jiangsu Province, China. The pseudo-color images of bands 33, 22, and 11 and the corresponding ground truth are shown in Fig. 4. Each image has 198 bands with a size of 463×241 pixels, and the spatial resolution was 30 m. In total, the dataset contains 9698 changed pixels and 101,886 unchanged pixels, which reflected the changes in the river caused by the removal of sediment.

The Farmland dataset was captured by the Hyperion sensor on the EO-1 satellite on May 3, 2006, and April 23, 2007, in Yancheng City, Jiangsu Province, China. The pseudo-color images of bands 33, 22, and 11 and the corresponding ground truth are shown in Fig. 5. Each image has 155 bands with a size of 450×140 pixels. In total, the dataset contains 18,277 changed pixels and 44,723 unchanged pixels, which reflected the changes in the farmland due to crop rotation.

The Hermiston dataset was collected by Hyperion sensor on May 1, 2004, and May 8, 2007, in the Hermiston City area (Oregon). The pseudo-color images of bands 33, 22, and 11 and the corresponding ground truth are shown in Fig. 6. Each image has 242 bands with a size of 390×200 pixels, and the spatial resolution was 30 m. In total, the dataset contains 9986 changed pixels and 68,014 unchanged pixels, which reflected the variations in the circular farmland area.

3.2. Experimental configuration

1. **Training and Testing Samples Split:** We randomly selected 2500 unchanged and 1250 changed pixels on the River dataset, accounting for 3.36% of the total for training. The 500 unchanged and 500 changed pixels on the Farmland dataset, accounting for

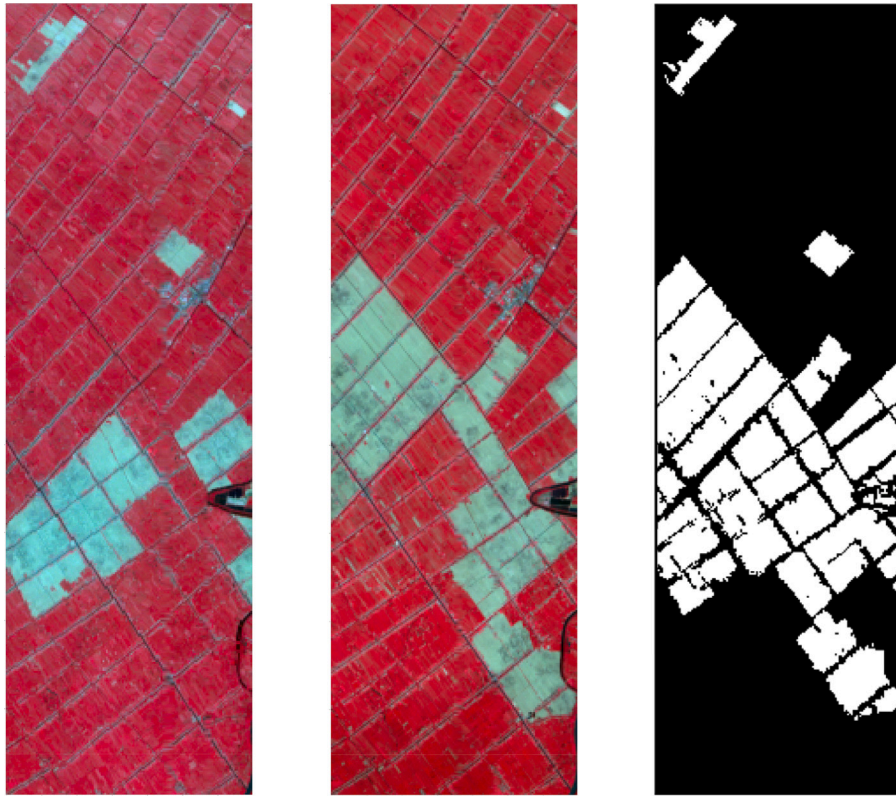


Fig. 5. Farmland dataset. From left to right: Image acquired on May 3, 2006; Image acquired on April 23, 2007; Ground Truth.

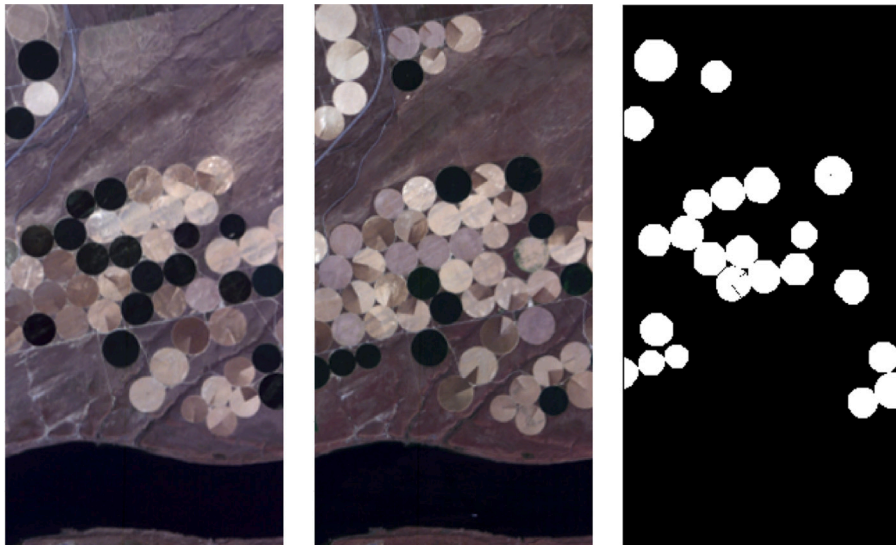


Fig. 6. Hermiston dataset. From left to right: Image acquired on May 1, 2004; Image acquired on May 8, 2007; Ground Truth.

Table 1

Number of training and test samples for three different datasets.

Dataset	Training Set			Test Set		
	Unchanged	Changed	Total	Unchanged	Changed	Total
River	2500	1250	3750	100 635	7198	107 833
Farmland	500	500	1000	44 223	17 777	62 000
Hermiston	5200	2600	7800	65 414	4786	70 200

1.58% of the total, were selected for training. We chose 5200 unchanged and 2600 changed pixels on the Hermiston dataset

for training, accounting for 10% of the total pixels that were selected for training. The training and test samples selection for three datasets are reported in Table 1.

2. *Evaluation Metrics:* We evaluate the proposed model in terms of three commonly-used indices, i.e., Overall Accuracy (OA), Kappa Coefficient (Kappa) and F1-score (F1). The detection maps obtained by different comparative methods are visualized for a qualitative comparison.
3. *Implementation Details:* We implemented our proposed method on the Pytorch 1.10.1 platform with Intel Core i9-10900k CPU and NVIDIA GeForce RTX 3060 12 G GPU. All the implementation

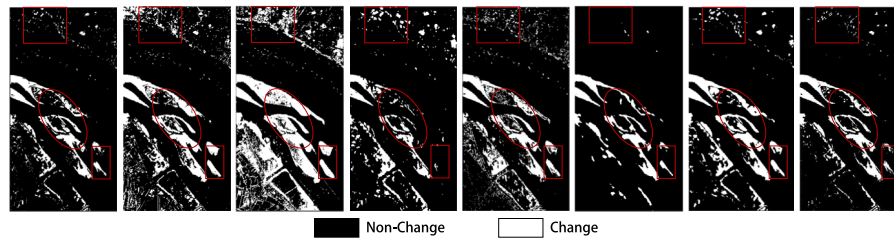


Fig. 7. The change detection maps on the River dataset. (a) Ground Truth (b) CVA (c) SVM (d) RSC (e) ML-EDAN (f) CSA (g) SST-Former (h) Proposed.

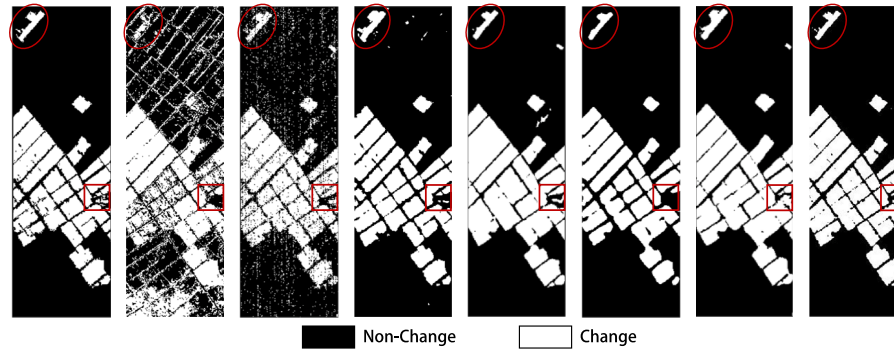


Fig. 8. The change detection maps on the Farmland dataset. (a) Ground Truth (b) CVA (c) SVM (d) RSC (e) ML-EDAN (f) CSA (g) SST-Former (h) Proposed.

were based on Python 3.7.0 language. Adam was adopted as the optimizer with a learning rate of 0.0005. The total epochs were empirically set to 900.

- For the CVA (Malila, 1980), the threshold selection is the most significant factor affecting the detection results. In our case, the OSTU (maximum interclass variance thresholding algorithm) unsupervised threshold segmentation algorithm is to be applied to select the threshold automatically.
- For the SVM (Nemmour & Chibani, 2006), the SVC package in scikit-learn 1.1.3 is selected for the implementation of HSI-CD tasks, which is performed by using radical basis function (RBF) kernel. The hyperparameters include the regularization parameter and tolerance for the stopping criterion, which is empirically set to 1000 and 0.001, respectively.
- For the RSCNet (Wang, Wang, et al., 2022), the architecture is based on the ResNet, which embedded multiple blocks of self-calibrated convolution. We refer to the original paper for more parameter configuration details about the model.
- For the ML-EDAN,¹ (Qu et al., 2021) the model is composed of two parallel architectures with multilevel encoder-decoder layers and LSTM network. We follow the original parameters configuration to obtain the optimal detection results.
- The CSANet² (Song et al., 2022) comprises a cross-temporal interaction symmetric attention (CSA) module to interact with the bi-temporal HS information. Same as our proposed DBS³TAN, CSANet also utilized self-attention for feature extraction. As a result, it is reasonable to choose the model for comparison. As for constructing the structure and selecting parameters.
- The SST-Former³ (Wang, Hong, et al., 2022) also employs the transformer-based structure for HSI-CD tasks. Accordingly, it is reasonable for us to select the model to compare the HSI-CD performances with our proposed DBS³TAN. Additionally, we refer to the original work to achieve the optimum results.

3.3. Discussion

(1) *Ablation Study*: To fully evaluate the proposed DBS³TAN, we investigate the performance in terms of OA by adding different modules in a stepwise fashion. The extensive ablation experiments are conducted on three datasets to verify the effectiveness of these components in the proposed DBS³TAN for HSI-CD, as listed in Table 2. As expected, the proposed spatial-spectral transformer (SST) without spatial attention or spectral transformer modules yields lower change detection results. To be specific, the results obtained without the SST module are better than the model without the spatial attention module (beyond around 0.5%–1% OAs on three datasets), which might indicate the SST has a more significant role in improving the detection performances compared with the spatial attention module to some extent. More importantly, combining the two modules can significantly boost performance (about 1%–2% OAs). The proposed DBS³TAN exhibits promising detection results on three datasets, and all the component modules play critical roles in spatial-spectral feature extraction. We also conduct additional ablation experiments to thoroughly analyze the SST module in improving the detection performances, especially for the module. As Section 2 mentions, the SST module consists of the two transformer encoders, i.e., spatial and spectral transformer, respectively. As a result, we investigate the performance of the SST on three datasets by adding the spatial transformer encoder and the spectral encoder in a stepwise fashion. Other parameter settings are consistent with the original model. Table 2 illustrates that both encoders have improved the final detection results to varying degrees. Specifically, the model without spectral encoder outperforms the comparison group by around 1% OAs on the River dataset. However, it performs worse by 1% OAs on the Farmland dataset. They exhibit similar detection results on the Hermiston dataset.

Additionally, we also conducted extensive experiments to assess the performances with depthwise separable convolutions (DSConv) and ordinary convolutions (ORDConv) in terms of OA, Kappa and F1 on three datasets. The results are shown in Fig. 11. It is observed that replacing DSConv with ORDConv resulted in a limited increase in three indicators' accuracies (i.e., OA 0.3%, Kappa 0.1%, F1 0.2%). Although ORDConv improved detection capability, it came with a substantial

¹ <https://github.com/Jiahuiqu/ML-EDAN>.

² <https://github.com/srxlnnu>.

³ https://github.com/yanhengwang-heu/IEEE_TGRS_SSTFormer

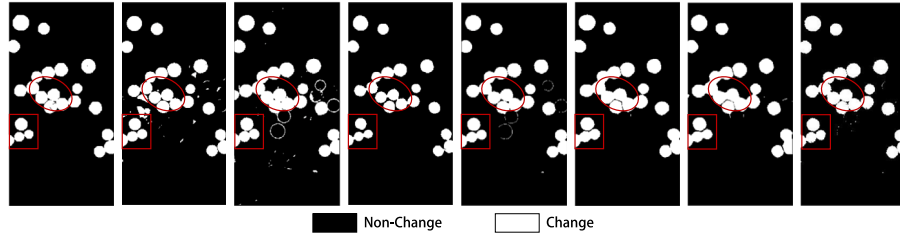


Fig. 9. The change detection maps on the Hermiston dataset. (a) Ground Truth (b)CVA (c) SVM (d) RSC (e) ML-EDAN (f) CSA (g)SST-Former (h) Proposed.

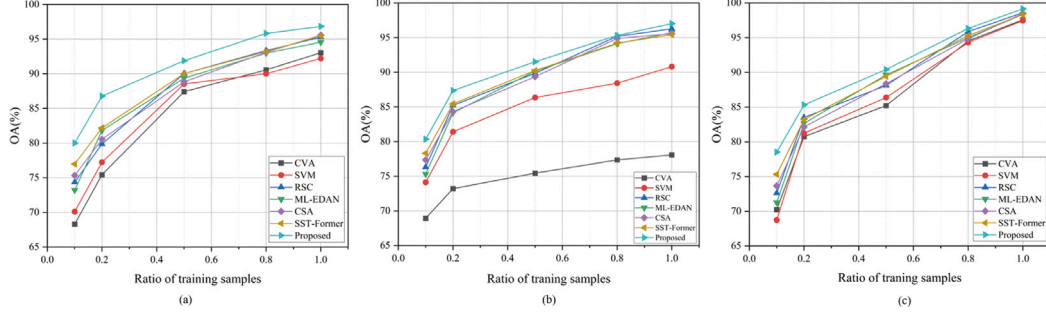


Fig. 10. The OA obtained by multiple models with different ratio of training samples for (a) River dataset, (b) Farmland dataset, (c) Hermiston dataset.

Table 2

Ablation analysis of the proposed model with combination of different modules on three datasets.

Datasets	Modules	Spatial-spectral transformer		Metrics		
		Spatial only	Spectral only	OA (%)	Kappa	F1 (%)
River	✓			94.01	0.7358	64.54
	✓	✓	✓	94.39	0.7417	68.79
	✓	✓		95.88	0.6387	71.82
	✓		✓	95.08	0.7540	75.71
	✓	✓	✓	96.84	0.7750	79.21
Farmland	✓			93.80	0.8978	88.74
	✓	✓	✓	96.70	0.8711	90.41
	✓	✓		95.82	0.9071	91.68
	✓		✓	96.73	0.9083	92.37
	✓	✓	✓	97.03	0.9278	94.87
Hermiston	✓			98.72	0.9167	87.14
	✓	✓	✓	98.97	0.9243	88.16
	✓	✓		98.85	0.9475	89.34
	✓		✓	98.83	0.9514	92.47
	✓	✓	✓	99.16	0.9559	96.06

computational cost and memory overhead. In comprehensive consideration, we selected DSConv to maintain the model lightweight while achieving better detection results.

(2) *Parameter Sensitivity Analysis*: Apart from the trainable parameters in the training process, some hyperparameters also play a vital role in the final detection performances. Specifically, the number of encoders, the attention heads, split image sizes and the threshold values between spatial and spectral branches all significantly influence the performances of HSI-CD. Therefore, it is essential that we investigate the hyperparameters range and select the optimal from the results. Similarly, we evaluate the sensitivity of parameters in an ablation manner, i.e., by only modifying the value of the corresponding hyperparameter. Fig. 12(a) illustrates the changing trend of the detection accuracies with the gradual increase of the attention heads in terms of OA. It can be seen that the OA values increase firstly with heads of spatial-spectral transformer increase from 1 to 2 and then decrease gradually with the number of heads increases again on three datasets. We may conclude that the number of heads enhances the multiscale feature extraction ability within a certain range. Excessive heads are likely to lead to the overfitting of the network. Such a theory seems to

apply to analyzing the number of transformer encoders. As seen from Fig. 12(b), the OA values peak at the number is 2 on the River and Hermiston dataset; for the Farmland, it peaks at 3. It is reasonable that due to the variability of datasets, the optimal results tend to be obtained at different hyperparameter values. For the threshold values between spatial and spectral branches, we select the multiple values, i.e. from 0.1 to 0.9 with intervals of 0.1, to conduct the experiments. It can be seen from Fig. 12(c) that the OA values increase sharply when the threshold values change from 0.1-0.5, and then fluctuate between 0.5 and 0.9. To further refine the detection results, we additionally conduct extensive experiments for the threshold values in the interval of 0.5-0.9 to select the optimal threshold values. As a result, the final threshold values are selected to 0.8, 0.99 and 0.75 for River, Hermiston and Farmland datasets, respectively. We also select image patch sizes 3×3 , 5×5 , 7×7 , 9×9 to analyze the impact on the proposed model. From Fig. 12(d), we can see the OA first increases when the patch size is from 3×3 to 5×5 and then decreases obviously. As a result, the patch size of 5×5 is the optimal choice for accurate HSI-CD. Moreover, to evaluate the influence of the weights factor on the HSI-CD performances, We have selected multiple values of λ ranging from 0.1

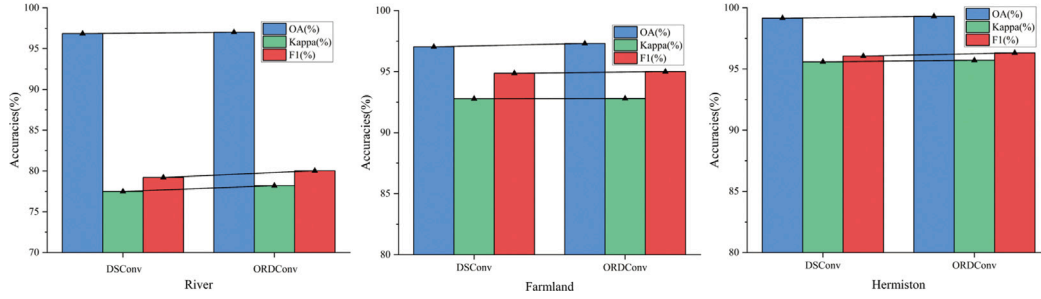


Fig. 11. The OA, Kappa and F1 on three datasets under DSConv and ORDCov.

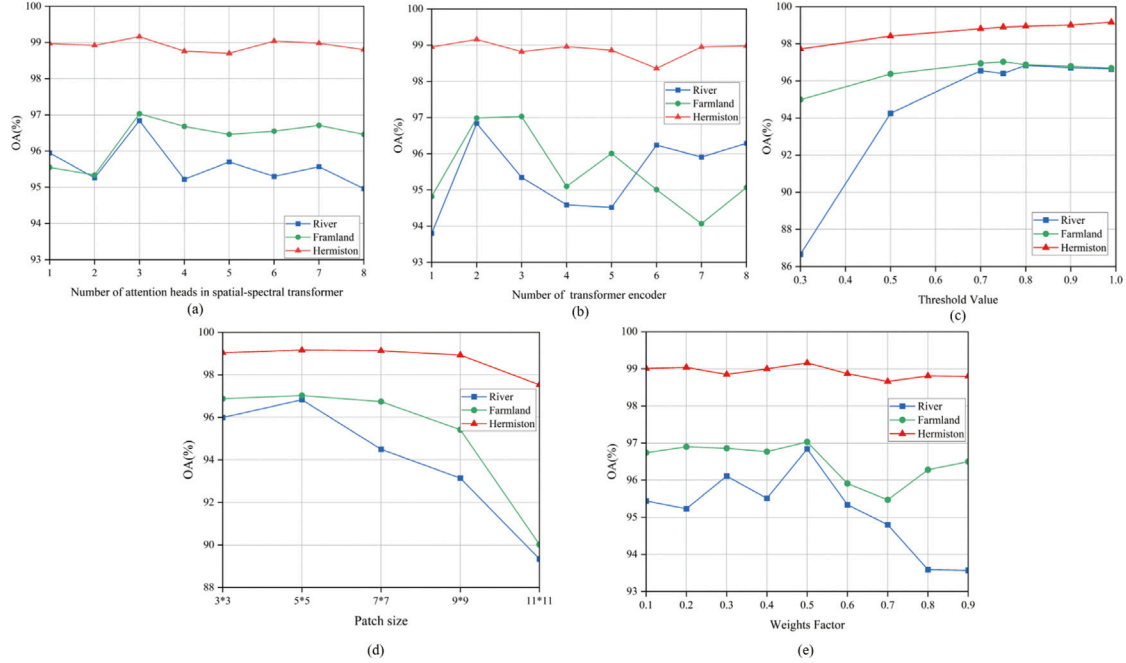


Fig. 12. Different parameter sensitivity analysis in terms of OA on three datasets; (a) the number of attention heads in spatial-spectral transformer; (b) the number of transformer encoder; (c) threshold value; (d) patch size; (e) weights factors.

to 0.9 with an interval of 0.1. Specifically, e.g., when $\lambda = 0.1$, the weight of the spatial attention branch is 0.1, and that of the spatial-spectral transformer branch is 0.9. The corresponding results are illustrated in Fig. 12(e). In summary, the most favorable experimental outcomes were achieved on all three datasets when the weight factor was set to 0.5.

(3) *Computational Complexity Analysis*: We selected four deep learning (DL)-based models as comparative groups: RSC, ML-EDAN, CSA, and SST-Former. We evaluated all models regarding network parameters and floating point operations per second (Flops). The results are reported in Table 4. Generally, transformer-based models tended to have higher time complexity than CNN-based models. The proposed DBS³TAN did not perform the best in terms of parameters and Flops, mainly due to the inherent structures of transformers. Nonetheless, it outperformed other comparative models regarding HSI-CD performances, making it highly competitive. Overall, the proposed DBS³TAN is still a strong contender in this domain.

(4) *Different Ratio of Training Samples Discussion*: To assess the performance of the proposed model when trained with limited samples, we select 10%, 30%, 50% and 80% as new training sets for experiments based on the original training samples. The results are illustrated in Fig. 10. Our proposed model also has great detection performance in the face of limited training samples, which is better than all the other methods on three datasets.

(5) *Analysis of the weighted contrastive loss function*: We compare the change detection performances between the proposed weighted contrastive loss and the cross-entropy loss to demonstrate the validity of the proposed weighted contrastive loss function. The results are shown in Fig. 13. Overall, the performance of using the weighted contrastive loss function is superior to that of the cross-entropy loss on three datasets in terms of OA, Kappa and F1. The reason for the phenomenon is that the weighted contrastive function is adopted to train the proposed network based on the appealing property that the feature tensors of the unchanged pixel pairs are close to each other and those of changed pixel pairs are far away.

3.4. Quantitative results and analysis

Table 3 presents the quantitative change detection results for the River, Farmland, and Hermiston datasets, as measured by the OA, Kappa, and F1-score indices. In general, conventional classifiers such as CVA and SVM achieve comparable detection performance on the River and Hermiston datasets, with the exception of the accuracies obtained by CVA on the Farmland dataset, which are significantly lower than those achieved by SVM. It is worth noting that DL-based methods, such as RSC, ML-EDAN, and CSA, perform better than the aforementioned classifiers due to their superior ability to learn powerful features. These results demonstrate the advantage of DL-based methods in HSI-CD to some extent. Furthermore, transformers such as SST-former can extract

Table 3

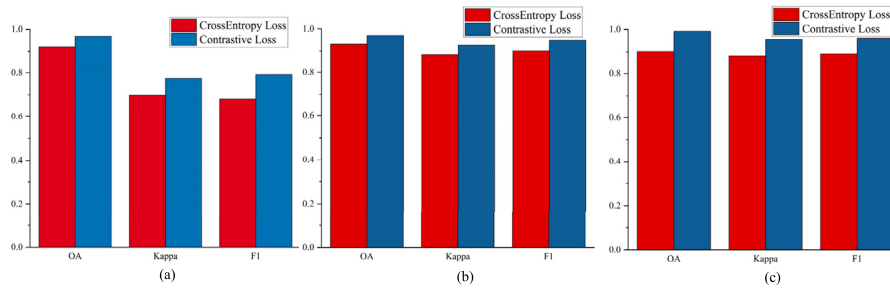
Quantitative performance of different detection methods in terms of OA, Kappa, and F1 on three datasets.

Datasets	Metric	CVA	VM	RSC	ML-EDAN	CSA	SST-Former	DBS ³ TAN
River	OA (%)	93.05	92.22	95.29	94.56	95.57	95.47	96.84
	Kappa	0.6691	0.5952	0.7347	0.6934	0.6829	0.7399	0.7750
	F1 (%)	75.54	63.53	76.39	72.65	70.63	76.43	79.21
Farmland	OA (%)	78.08	90.81	96.28	95.73	95.57	95.44	97.03
	Kappa	0.5407	0.7829	0.909	0.8989	0.8899	0.8925	0.9278
	F1 (%)	70.22	84.86	94.40	93.94	92.05	93.72	94.87
Hermiston	OA (%)	97.58	97.46	98.6	98.36	98.49	98.33	99.16
	Kappa	0.8944	0.8281	0.9215	0.9172	0.9311	0.9163	0.9559
	F1 (%)	90.83	84.16	93.22	93.08	94.03	92.39	96.06

Table 4

Parameters and Flops for several DL-based models on three datasets.

	Indicators	RSC	ML-EDAN	CSA	SST-Former	DBS ³ TAN
River	Parameter (M)	0.03	88.53	2.45	120.78	82.15
	Flops (G)	0.01	18.24	2.31	56.4	25.31
Farmland	Parameter	0.02	88.13	2.43	118.62	80.53
	Flops (G)	0.01	17.6	2.24	54.2	26.21
Hermiston	Parameter	0.03	88.93	2.48	121.6	83.12
	Flops (G)	0.01	18.89	2.37	48.2	24.12

**Fig. 13.** The OA, Kappa and F1 on three datasets under cross-entropy and weighted contrastive loss; (a) River dataset, (b) Farmland dataset, (c) Hermiston dataset.

fine spectral representations from a sequence perspective, yielding comparable performance to CNN-based models. Although the SST-Former exhibits powerful abilities in modeling global sequential information, its ability to model local spatial discrepancies is limited. The proposed DBS³TAN provides new inspiration by extracting local spatial and global spectral features using a two-branch siamese network, resulting in even superior detection performance than state-of-the-art models.

3.5. Visual evaluation

To better compare the detection performances obtained by all methods, we conducted a qualitative evaluation by visualizing the detection maps, as shown in Figs. 7–9. The obtained results for the River, Farmland, and Hermiston datasets reveal that conventional detection methods, such as CVA, SVM, and MLP, tend to generate detection maps with a high degree of salt and pepper noise, indicating that these classifiers struggle to accurately identify areas of change and areas that remain unchanged. In contrast, DL-based models, such as ML-EDAN and CSA, can produce smoother detection maps due to their powerful feature representation capabilities. SST-Former, as an emerging architecture, can extract sequential representations from multiple temporal HSIs, resulting in visualized detection maps comparable to those obtained by other backbone models. By deeply mining hidden features in spatial and spectral dimensions using self-attention mechanisms and transformer structures, the proposed DBS³TAN achieves perfect detection maps on all three datasets. Furthermore, the regions of interest (ROIs) highlighted by the red borderlines in the figures highlight differences in the detection maps between different models, facilitating intuitive performance evaluation. As can be seen from these ROIs, the proposed

model exhibits clearer and finer details about change detection boundaries. The detection maps of DBS³TAN on all three datasets contain far less salt and pepper noise than other comparative models, which is consistent with the results of the quantitative analysis.

4. Conclusions

In this article, a dual-branch siamese spatial-spectral transformer attention network (DBS³TAN) was proposed for accurate detection of land cover changes in bi-temporal HSIs. Considering the advantages of CNN and transformers, we combine them in two branches for spatial and spectral feature extraction, respectively. Furthermore, we improve the standard transformer structure, which not only focuses on the sequence attributes of spectral signatures commonly but also mines the spatial features from the relationship between adjacent pixels. The designed dual-branch siamese network was trained by the weighted contrastive loss function, which assigned different weights to balance the contributions of two branches, accelerating the convergence of the network simultaneously. Experimental results on three HSI datasets demonstrated that the proposed model performs better than other comparison methods.

CRedit authorship contribution statement

Yiyan Zhang: Conceptualization, Methodology, Software. **Tingting Wang:** Data curation, Writing – original draft. **Chenkai Zhang:** Visualization. **Shufang Xu:** Supervision, Reviewing. **Hongmin Gao:** Validation, Reviewing. **Chenming Li:** Investigation, Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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